53

DEPARTMENT OF MECHANICAL ENGINEERING
M B M UNIVERSITY , JODHPUR

B.E. II YEAR III SEMESTER
Second Mid Term Examination
(SESSION: 2025-26)

Subject: Mathematics 3 ME 21 A (MA) M.M :10

Solve any four questions. (Each Carry Equal Marks) Time: 1 Hour

Q.1 Use Milne's Predictor – Corrector method to solve the equation

$$\frac{dy}{dx} = x - y^2$$

At x = 0.8 given that y(0) = 0, y(0.2) = 0.02, y(0.4) = 0.0795, y(0.6) = 0.1762

Q.2 Use Stirling formula to find y_{28} Given that

$y_{20} = 49225$, $y_{25} = 48316$, $y_{30} = 47236$, $y_{35} = 45926$, $y_{40} = 44306$

Q.3 Using Lagranges interpolation formula to find the value of y when x = 3 from the following data

x	0	1	4	5
y	3	4	24	39

Q.4 Solve by Gauss Elimination Method

$x - y + 2z = 3$

$x + 2y + 3z = 5$

$3x - 4y - 5z = 13$

Q.5 Find f (1934) by Newton's Forward Interpolation Formula.

Year x	1931	1941	1951	1961	1971	1981
Sale f (x)	12	15	20	27	39	52

2nd Midterm

ME 201A: Engineering Thermodynamics

Date 07/11/2025 Exam Duration: One Hour (2:30 PM to 3:3 PM) M.M.:20

CO3	1.	Write Vander Waals equation for real gases. Explain Dalton's law of partial pressure and Gibbs theorem.**[2+2+2]**
CO5	2.	Write Maxwell's four equations (in differential form). **[5]**
CO5	3.	What is throttling process, joule kelvin coefficient and inversion curve? Plot T-P diagram showing inversion curve. [4]
CO6	4.	A two stage air compressor with perfect intercooling takes in air at 1 bar pressure and 27°C. The law of compression in both stages is $pv^{1.3}$ = constant. The compressed air is delivered at 9bar from the HP cylinder to an air receiver. Calculate per kilogram of air, the minimum work done, and the heat rejected to the intercooler. **[5]**

MBM University, Jodhpur
Department of Mechanical Engineering
B.E. IInd Year IIIrd Semester
Second Mid-Term Examination Session 2025-26
3ME43A: Kinematics of Machines

Time: 1 hour **Maximum Marks – 10**

Q. 1(a). PQRS is a four-bar chain with link PS fixed. The lengths of the links are PQ = 70mm, QR = 180 mm; RS = 110 mm and PS = 200 mm. The crank PQ rotates at 20 rad/sec clockwise. Draw the velocity and acceleration diagram when angle QPS = 60° and Q and R lie on the same side of PS. Find the linear and angular velocity of QR and RS. (5)

Q. 2 (a) Sketch a pantograph, explain its working and show that it can be used to reproduce to an enlarged and reduced scale a given figure. (3)

(b) What is a Hooke's joint? Where is it used? (2)

DEPARTMENT OF MECHANICAL ENGINEERING

MBM UNIVERSITY, JODHPUR

BE II Yr. III SEM.

Material Technology

Paper Code- 3ME42A(ME)

Mid Term Exam.-II

Date: 08/11/25 Max. Marks: 10

All questions compulsory

Q.1	CO2	Describe the Recovery and Recrystallization process with details.	3
Q.2	CO2	Describe the effect of temperature on different mechanical properties.	2
Q.3	CO2	Describe the Iron-Iron carbon diagram with diagram and details.	5

MID TERM II (B.E. III SEM 2025-26)

3ME44A(ME) Mechanics of Solids

DEPT. OF MECH.ENGG., MBM UNIVERSITY

ME: 60 MIN MM:10

) III Q1. An object is subjected to two perpendicular stresses 80 N/mm^2 and 20 N/mm^2 tensile and 40N/mm^2 of ear stress. Using Mohr's circle method find σ_t, σ_n and σ_r on a plane inclined at 30° to normal cross-section. (4)

) IV Q2. Derive Euler's formula for column hinged at both the ends. (3)

) V Q3. A beam of length 10 m is simply supported at its ends carries two point loads of 100 kN and 60 kN a distance of 2 m and 5 m respectively from the left support. Calculate the deflection under each load. Find so the maximum deflection. Take $E = 2 \times 10^5$ N/mm^2 and $I = 18 \times 10^8$ mm^4. (3)